



Nebraska Public Power District

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NLS2020003
February 4, 2020

U.S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, D.C. 20555-0001

Subject: Licensee Event Report No. 2019-003-00
Cooper Nuclear Station, Docket No. 50-298, DPR-46

Dear Sir or Madam:

The purpose of this correspondence is to forward Licensee Event Report 2019-003-00.

There are no new commitments contained in this letter.

Sincerely,

John Dent, Jr.
Vice President -
Chief Nuclear Officer

/tf

Attachment: Licensee Event Report 2019-003-00

cc: Regional Administrator w/attachment
USNRC - Region IV

NPG Distribution w/attachment

Cooper Project Manager w/attachment
USNRC - NRR Plant Licensing Branch IV

INPO Records Center w/attachment
via ICES entry

Senior Resident Inspector w/attachment
USNRC - CNS

SORC Chairman w/attachment

SRAB Administrator w/attachment

CNS Records w/attachment

IE22
NRR



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form <http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name Cooper Nuclear Station	2. Docket Number 05000298	3. Page 1 of 5
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4. Title**Division 2 Service Water Discharge Blockage Resulting in Unplanned Service Water Inoperability**

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
12	06	2019	2019	- 003	- 00	02	04	2020	Facility Name	Docket Number
										05000
									Facility Name	Docket Number
										05000

9. Operating Mode 1	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
10. Power Level 100	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
☐ 50.73(a)(2)(i)(C) ☐ Other (Specify in Abstract below or in NRC Form 366A)				

12. Licensee Contact for this LER

Licensee Contact

Linda Dewhirst, Regulatory Affairs and Compliance Manager

Telephone Number (Include Area Code)

(402) 825-5416

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To ICES	Cause	System	Component	Manufacturer	Reportable To ICES

14. Supplemental Report Expected☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No**15. Expected Submission Date**

Month	Day	Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On December 5, 2019, while attempting to place Reactor Equipment Cooling Heat Exchanger B in service, Service Water flow to the heat exchanger was unable to be established. Reactor Equipment Cooling Subsystem B was declared Inoperable at 02:15 on December 6, 2019 following validation of the condition. On December 8, 2019, troubleshooting determined the flow blockage was on the common buried Division 2 Service Water piping downstream of Diesel Generator 2, Reactor Equipment Cooling Heat Exchanger B, and Residual Heat Removal Heat Exchanger B. Service Water Subsystem B and Diesel Generator 2 were declared Inoperable at 05:00 on December 8, 2019.

Manual sounding of the discharge canal was performed to determine the sediment levels above the Division 2 Service Water pipe outfall. The sounding identified approximately 15 feet of sediment. A Work Order was prepared for removing the sediment by dredging the discharge canal. Dredging commenced on December 12, 2019. Within one hour, Service Water flow was established through the Division 2 outfall. Dredging continued to remove sediment from around the Division 2 outfall, then proceeded to remove sediment from around the Division 1 outfall. A Compensatory Measure was implemented to maintain continuous Service Water flow from the outfalls above 1000 gallons per minute (gpm). Operability was restored at 05:00 on December 13, 2019.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME		2. DOCKET NUMBER		3. LER NUMBER		
Cooper Nuclear Station		05000298		YEAR	SEQUENTIAL NUMBER	REV NO.
				2019	- 003	- 00

NARRATIVE**PLANT STATUS**

Cooper Nuclear Station was in Mode 1, Power Operation, at 100 percent power, on December 6, 2019, when the event was discovered. At the start of the event, Service Water Pump A was inoperable for planned maintenance.

BACKGROUND

The Service Water System is designed to provide cooling water for the removal of heat from equipment, such as the diesel generators [EIS: EK] and Reactor Equipment Cooling System heat exchangers [EIS: CC], and to provide a supply of water for the Residual Heat Removal-Service Water Heat Exchangers [EIS: BO] through the Residual Heat Removal-Service Water Booster System pumps [EIS: BO], required for a safe reactor shutdown following a Design Basis Accident or transient. The Service Water System also provides cooling to unit components, as required, during normal operation. The Service Water System also provides cooling water to turbine building non-essential loads [EIS: KB].

The Service Water System consists of the Ultimate Heat Sink and two independent and redundant subsystems. Each of the two Service Water subsystems is made up of a header, two 8000 gpm pumps, a suction source, valves, piping and associated instrumentation. Either of the two subsystems is capable of providing the required cooling capacity to support the required systems with one pump operating. The two subsystems are separated from each other so failure of one subsystem will not affect the Operability of the other system.

Cooling water is pumped from the Missouri River by the Service Water pumps to the essential components through the two main headers. After removing heat from the components, the water is collected into two discharge headers and routed to the discharge canal where the water is returned to the river. Service Water discharge from the turbine equipment cooling heat exchangers is routed to the circulating water discharge tunnel.

EVENT DESCRIPTION

On December 5, 2019 at 23:40, Operations personnel attempted to place Reactor Equipment Cooling Heat Exchanger B in service to support the performance of post maintenance testing for Service Water Pump A. Service Water flow to Reactor Equipment Cooling Heat Exchanger B was established to match the flow through Reactor Equipment Cooling Heat Exchanger A at approximately 1700 gpm at which time Service Water flow to Reactor Equipment Cooling Heat Exchanger A was secured by closing the Service Water Reactor Equipment Cooling Heat Exchanger A outlet valve. At that time, Service Water flow to Reactor Equipment Cooling Heat Exchanger B was noticed to be 0 gpm. The Service Water Reactor Equipment Cooling Heat Exchanger B outlet valve was opened an additional 10 seconds with no response noted. Reactor Equipment Cooling temperature began to rise at which time Operations personnel placed Reactor Equipment Cooling Heat Exchanger A back in service.

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The condition was validated on December 6, 2019 at 00:54 by opening the Service Water Reactor Equipment Cooling Heat Exchanger B outlet valve, which briefly established an indicated flow of 2400 gpm but then dropped to 0 gpm. The Reactor Equipment Cooling Heat Exchanger B Service Water outlet valve was fully opened with no system response. Technical Specification (TS) Limiting Condition of Operation (LCO) 3.7.3 Condition B was entered for Reactor Equipment Cooling Subsystem B being declared Inoperable on December 6, 2019 at 02:15.

Troubleshooting instructions were developed to confirm that the flow indication was valid, various Service Water valves did not have stem disc separation, or Reactor Equipment Cooling heat exchanger blockage was preventing flow. On December 7, 2019 at 23:30, Revision 2 of the troubleshooting installed pressure gauges on various system vents to confirm the location of the blockage. This troubleshooting revision confirmed the blockage was downstream of the last manual valve following of Reactor Equipment Cooling Heat Exchanger B. On December 8, 2019, Revision 3 of the troubleshooting determined the flow blockage was on the common buried Division 2 Service Water piping downstream of Diesel Generator 2, Reactor Equipment Cooling Heat Exchanger B, and Residual Heat Removal Heat Exchanger B. TS LCO 3.7.2 Condition A for the Service Water subsystem B inoperability and 3.8.1 Condition B for the Diesel Generator 2 inoperability were entered on December 8, 2019 at 05:00.

A Division 1 Diesel Generator Service Water Check Valve and Sump Test was completed on December 8, 2019 at 15:15 to ensure that Diesel Generator 1 was not Inoperable to due to a common cause failure as required by TS LCO 3.8.1 Condition B.3.1. On December 9, 2019, manual sounding of the discharge canal was performed to determine the sediment levels above the Division 2 Service Water pipe outfall. The sounding identified approximately 15 feet of sediment in the discharge canal in the area of the Service Water outfalls. A Work Order was then planned to remove the sediment by dredging the discharge canal with a barge and pump.

A Temporary Configuration Change was developed to route Service Water from downstream of Diesel Generator 2 back to the Intake Structure, if necessary, during a plant event. The Temporary Configuration Change was installed and Diesel Generator 2 was made available on December 11, 2019 at 05:21. Similarly, a Temporary Configuration Change was prepared to provide an alternate flow path from Division 2 Residual Heat Removal-Service Water to allow its restoration to an available status. However, Service Water Division 2 flow was restored prior to approval or installation of the Division 2 Residual Heat Removal-Service Water Temporary Configuration Change.

The barge arrived at the mouth of discharge canal on December 11, 2019. Service Water Booster Pump A was started to raise flow from the Division 1 Service Water outfall as a precautionary action during dredging activities. The barge maneuvered into position and commenced dredging on December 12, 2019 at 02:32. At 03:20, Service Water flow of 2000 gpm was established through Reactor Equipment Cooling Heat Exchanger B. Operations then adjusted Service Water flow through Reactor Equipment Cooling Heat Exchanger B to 3000 gpm. Service Water Booster Pump B was started to raise total Service Water flow from the Division 2 outfall to approximately 7000 gpm while dredging continued around the Division 2 outfall.

All flow was secured from the Division 2 outfall on December 12, 2019 at 11:36 to remove the implemented Temporary Configuration Change. Diesel Generator 2 was restored to a standby status on December 12,

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2019 at 21:35 and Post Maintenance Testing was completed on December 13, 2019 at 04:30. Service Water flow through Reactor Equipment Cooling Heat Exchanger B was established at 1500 gpm on December 13, 2019 at 04:33 as a Compensatory Measure to maintain Service Water Operability.

TS LCO 3.7.2 Condition A, TS LCO 3.7.3 Condition B, and TS LCO 3.8.1 Condition B were exited at 05:00 on December 13, 2019. Service Water Subsystems A and B were declared Operable-Comp Measure. A Compensatory Measure has been implemented to maintain continuous Service Water flow from the outfalls above 1000 gpm.

BASIS FOR REPORT

A review of past operability identified that between December 4, 2019 at 22:30 and December 4, 2019 at 22:33, and between December 5, 2019 at 02:40 and December 7, 2019 at 19:39, both divisions of Service Water were inoperable. Inoperability of both divisions of Service Water represents a condition that could have prevented fulfillment of a safety function per the guidance of NUREG 1022, Revision 3. This Inoperability lasted for greater than the TS LCO 3.7.2 Condition B Completion Time for Both Service Water Systems Inoperable and thus represents a condition prohibited by Technical Specifications for a failure to carry out the required actions of LCO 3.7.2 Condition B. Additionally, since Inoperability of Division 2 Service Water results in entry into TS LCO 3.8.1 for the corresponding Division 2 Diesel Generator, and the past operability review identified firm evidence that the condition could have existed since December 4, 2019 at 08:41, this also represents a failure to meet the required actions and completion times for TS LCO 3.8.1 Condition B.

A 60 Day Licensee Event Report is required by 10 CFR 50.73(a)(2)(v)(B) and (D) for a condition that could have prevented fulfillment of a safety function for the Service Water System, 10 CFR 50.73(a)(2)(i)(B) for a condition prohibited by Technical Specifications for the failure to meet the required actions and completion times of TS LCO 3.7.2 Condition B, and 10 CFR 50.73(a)(2)(i)(B) for a condition prohibited by Technical Specifications for the failure to meet the required actions and completion times of TS LCO 3.8.1 Condition B.

SAFETY SIGNIFICANCE

A sediment blockage at the Division 2 Service Water discharge prevented flow from being supplied through the Reactor Equipment Cooling Heat Exchanger B, Diesel Generator 2, and Residual Heat Removal Heat Exchanger B. Residual Heat Removal Heat Exchanger B and Diesel Generator 2 were in standby status for the duration of the event and were not needed in respond to an actual plant condition. The B Reactor Equipment Cooling Heat Exchanger was not required to support normal or emergency plant operations for the duration of the event. Throughout this event, at least one Service Water Pump in Division 1 was available and in operation. Although during portions of the event the Division 1 Service Water subsystem was inoperable, the system was capable of performing its required safety function with a single pump. As a result, there was no actual loss of safety function and no actual consequences to general safety of the public, nuclear safety, industrial safety, or radiological safety for this event. As such, this event will not be counted as a Safety System Functional Failure for the Nuclear Regulatory Commission performance indicator since no loss of safety function occurred.

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The ability to provide onsite emergency AC power is dependent on the ability of the Service Water System to cool the Diesel Generators. The long-term cooling capability of the Residual Heat Removal, Core Spray and Residual Heat Removal-Service Water Booster pumps is dependent on the cooling provided by the Service Water system. The potential consequence to general safety of the public, nuclear safety, industrial safety and radiological safety of this event, if the Division 1 Service Water subsystem failed during the Division 2 flow blockage, would be a complete loss of the Ultimate Heat Sink.

Actions taken to reduce the safety significance during the event were:

- Immediate protection of Division 1 Service Water and associated components to reduce the potential for an inadvertent human performance error affect that subsystem.
- Implementation of a Temporary Configuration Change to provide an alternate Service Water discharge flow path for Diesel Generator 2 restored the diesel generator to an available status.
- Development of a Temporary Configuration Change to provide an alternate flow path from Division 2 Residual Heat Removal-Service Water, which would have allowed the restoration of the Division 2 Residual Heat Removal subsystem to an available status.

CAUSE

The direct cause is sediment accumulation in the discharge canal to an extent that the pressure applied by the Service Water system could not displace it. The root cause is the failure to recognize that separation of the divisional Service Water discharge piping (completed via modification in 2014) introduced a new failure mode, sediment blockage in the discharge canal, under historically high and prolonged river level conditions.

CORRECTIVE ACTIONS

A Compensatory Measure to maintain continuous Service Water flow from the outfalls greater than 1000 gpm will remain in place until a permanent change is implemented.

Cooper Nuclear Station will perform an Engineering Evaluation to change the system design to maintain Service Water flow from both divisional outfalls to prevent sediment blockage. This will allow the Compensatory Measure to be replaced by permanent plant design.

Corrective Actions to address the Extent of Cause include reviewing the design change implemented in 2014 to separate the Service Water Discharge piping to determine if other failure modes were not recognized, and reviewing other design changes to determine if sediment blockage was appropriately considered as a potential failure mode.

PREVIOUS EVENTS

Since plant startup there have been no previous events related to Service Water system discharge blockage.